Adding USGS channel measurements to RiverShapeML

Andrew Sommerlot · andrewsommerlot@gmail.com · November 2026

RiverShapeML (Modaresi Rad et al., 2024) predicts bankfull and in-channel width and depth for all 2.65 million NHDPlus reaches in CONUS from a model trained on HYDRoSWOT, the USGS collection of acoustic Doppler (ADCP) measurements through 2014. USGS now publishes every channel measurement it makes, by every method (wading, bridge, cableway, boat), through its Water Data OGC API: 4.77 million records. This note asks whether those measurements improve the model.

How it started. The USGS gage on Shoal Creek at Silverway Drive in Austin (08156675), at the end of a reach I walked, has 68 width measurements. HYDRoSWOT has 2, both from 2009, so the site falls below the 5-measurement minimum and is not used. The other 66, mostly wading measurements, are simply outside HYDRoSWOT. Across CONUS, that pattern turns out to be common, and most common in small streams.

What I did. I rebuilt the training targets from all USGS measurements, using RiverShapeML's sanity rules and Mike Johnson's AHGestimation for the hydraulic-geometry fits. Each site's curves are read at two flows from its annual peaks since water year 2000: bankfull at the 2-year flood (the median annual peak) and in-channel at the 1-year flood (the smallest annual peak). I replaced the $R^2 \geq 0.6$ site filter with a filter on the precision of the target itself (many well-measured sites have nearly constant width and fail R^2 for that reason alone). The features, feature engineering and inference are those of the public NOAA-OWP/predict-riverML pipeline, whose national outputs I could reproduce exactly from the published models; the model is the same XGBoost setup with modest regularization and early stopping.

More sites, especially in small streams. Bankfull-width training sites rise from 4,375 to 7,221 (+65%). Sites draining under 10 km² rise from 66 to 125, and 10–100 km² from 596 to 1,274; 58% of NHDPlus reaches are under 10 km². In-channel targets follow the same pattern (7,040 sites for width). At the same sites, bankfull widths from all measurements and from HYDRoSWOT alone agree at the median (ratio 1.00), and depths measured the same way agree within 0.1–2.4%.

Lower error in every size class, most in small streams and in in-channel depth. I trained the same model twice for each of the four quantities: on the existing RiverShapeML data (HYDRoSWOT measurements at existing sites) and on the expanded set (all measurements, existing and new sites), and scored both on the same held-out sites with cross-validation grouped by reach.

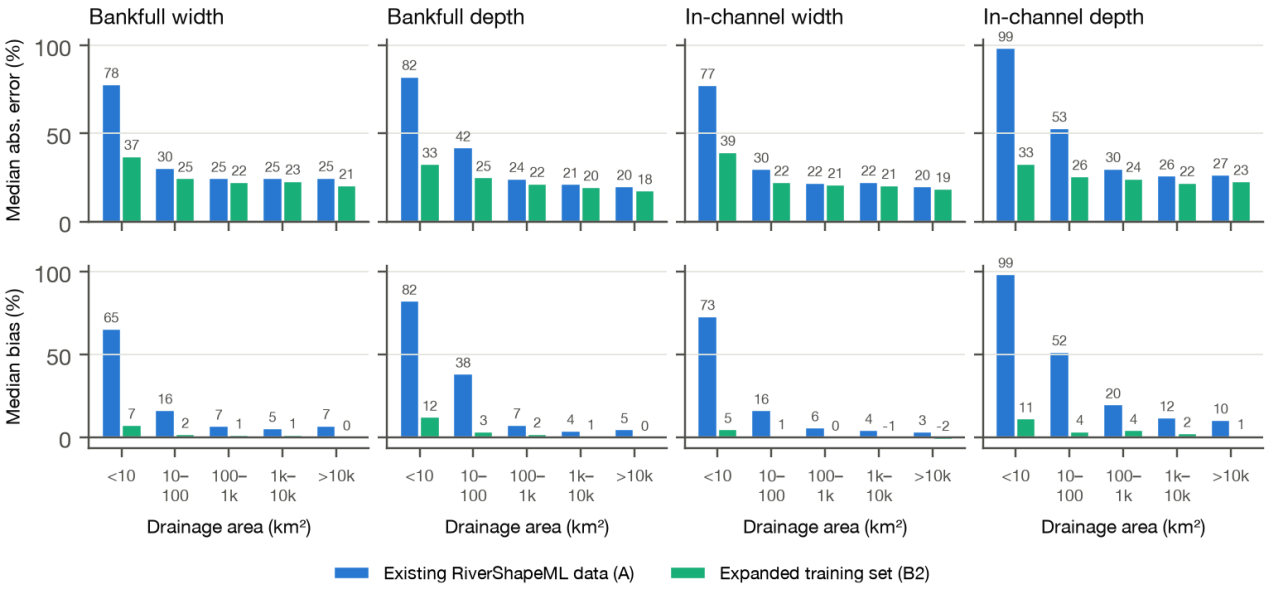


Figure 1. Held-out median absolute error and median bias by drainage area, existing RiverShapeML data (A) vs the expanded training set (B2), for bankfull and in-channel width and depth.

Median absolute error, A → B2	Bankfull width	Bankfull depth	In-channel width	In-channel depth
All sites	25.9% → 22.9%	24.9% → 21.2%	23.5% → 21.0%	31.2% → 23.8%
< 10 km²	78% → 37%	82% → 33%	77% → 39%	99% → 33%
Median bias, < 10 km²	+65% → +7%	+82% → +12%	+73% → +5%	+99% → +11%
Sites new to training	32% → 26%	42% → 25%	32% → 25%	56% → 28%
Existing sites, HYDRoSWOT-only targets	20.5% → 20.1%	18.4% → 18.4%	17.9% → 17.5%	20.9% → 20.7%

The existing-data model overpredicts small streams because it has few to learn from, and the effect is largest for in-channel depth, where the added wading measurements are densest. The gain comes from the new sites: at existing sites, judged on the original HYDRoSWOT targets, the two models match.

A re-predicted national product. I ran the retrained models through the predict-riverML pipeline for every reach, with split-conformal 90% prediction intervals by drainage-area class (coverage 0.88–0.90 on held-out sites for all four quantities). For bankfull, between 100 and 10,000 km² it agrees with the current model within about 10%. In headwaters it is narrower and closer to the Bieger et al. (2015) regional curves, so the national median width falls from 18.4 m to 10.7 m. Implausibly wide reaches (> 1 km) drop from 0.47% to 0.002%. In-channel width is 75% of bankfull width at the median, and in-channel depth 67% of bankfull depth. Above about 1,000 km² the in-channel values agree with the 2024 release; in small streams they are much smaller (median width under 10 km² 7.0 m against 22.6 m), partly because the in-channel flow is defined here as the smallest annual peak since 2000.

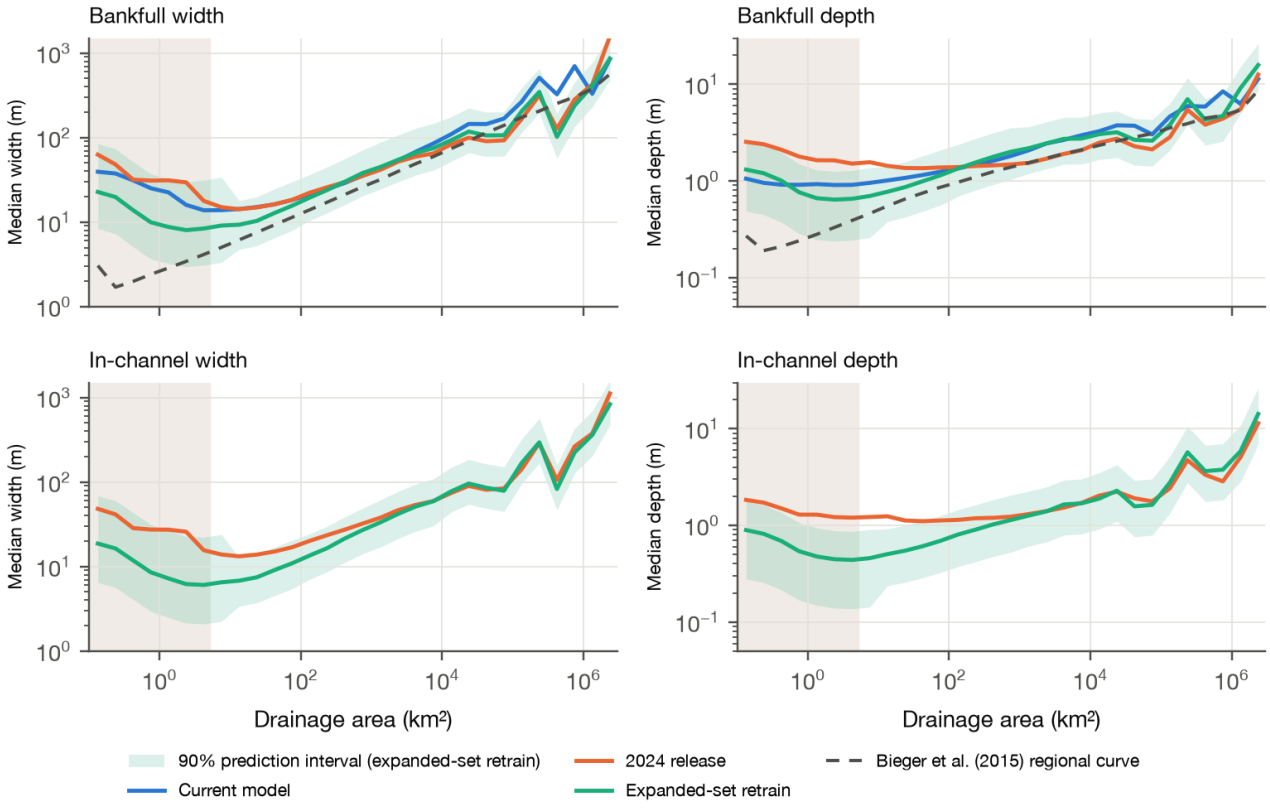


Figure 2. Median bankfull (top) and in-channel (bottom) width and depth by drainage area, all CONUS reaches. The current model has no in-channel output and the regional curves are bankfull only. The shaded region holds 44% of reaches but only ~55 training sites; the gap to the regional curves narrows but does not close.

Questions and discussion.

1. What was the original motivation for training on HYDRoSWOT, for example a single, consistent measurement method? The agreement above suggests data from other methods could be added.
2. About 7,000 more sites, mostly small streams, have enough measurements but short flow records. RiverShapeML's existing NWM flood-frequency option could be used to bring them in.
3. Which in-channel flow do downstream users, such as flood mapping, need? The added measurements change in-channel geometry most, so the definition matters most there.

Credit and citations. This builds directly on Arash Modaresi Rad's RiverShapeML and the NOAA Office of Water Prediction's open predict-riverML pipeline, whose published models and inference code made a controlled comparison possible, and on AHGestimation by Mike Johnson, with contributions from Arash Modaresi Rad. The code (one script rebuilds everything from public sources), a longer write-up and an interactive notebook are at github.com/northlinehydrology/rivershapeml-more-data; the re-predicted product, bankfull and in-channel width and depth with intervals for every reach, is in the v2.0 release.

References. Modaresi Rad, A., et al. (2024), *JGR: Machine Learning and Computation*, doi:10.1029/2024JH000173. Bieger, K., et al. (2015), *JAWRA* 51(3), 842–858, doi:10.1111/jawr.12282. Johnson, J. M., et al., AHGestimation (R package).